

AI-Enhanced Sensorless Control of Permanent Magnet Synchronous Motors (PMSM) for High-Performance Industrial Drives

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ABSTRACT

Permanent Magnet Synchronous Motors (PMSMs) have a high efficiency, power density, and compact size, and are increasingly popular in industrial automation, robotics, and electric vehicles, although sensorless control continues to be challenged by sensitivity to noise, parameter dependence and low-speed error under load perturbation. Recent AI-based technique can provide some solutions but is not strong enough to work in harsh industrial conditions. To solve that, an AI-enhanced sensorless control scheme is suggested, which combines a Convolutional Recurrent Neural Network (CRNN) to predict nonlinear states with an adaptive Sliding Mode Observer (SMO) to compensate the errors and improve the stability. The CRNN processes stator voltages and currents to estimate rotor dynamics and the SMO reduces parameter variability, noise and transients. The presented architecture is verified by the MATLAB/Simulink simulation and a hardware-in-the loop (HIL) test on a dSPACE real-time controller at low-speed, abrupt load and parameter variation conditions. Findings exhibit high estimation accuracy, reduction of torque ripple and rejection of disturbance over standalone EKF, SMO and pure AI estimation. These results validate the industrial relevance of the suggested hybrid design, and seasoned, energy efficient PMSM propulsion in the smart manufacturing, robotics, and environmentally friendly mobility projects.

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INTRODUCTION

Permanent Magnet Synchronous Motors (PMSMs) are becoming more predominant in industrial automation, robotics, electric vehicles (EVs) and aerospace systems because they are highly efficient, compact in size, and have high torque density compared to induction and brushed DC motors.^[1, 2] These have been used in the application where high precision and reliability is needed.^[3, 4] PMSM wants demand that is precise in

the position and speed of rotors, which is typically obtained via mechanical devices like encoders and resolvers. These sensors are precise, but they are quite costly and bulky and likely to fail when used in aggressive environments.^[6] These disadvantages are removed through sensorless control which does not use physical sensors.^[7] Nevertheless, traditional techniques, including back-EMF estimation, flux-linkage observers, Extended Kalman Filters (EKF), and

Sliding Mode Observers (SMO), cannot operate in low-speed mode because of low back-EMF amplitudes and high sensitivity to noise.^[8, 9] They also do not perform well at changes in the parameters (e.g., stator resistance) and during sudden changes in loads typical of industrial practice,^[10] which restricts their use in safety-critical applications such as traction drives and robotic actuators.^[11, 21] The recent development of Artificial Intelligence (AI) has resulted in new possibilities to control PMSM. Deep learning models such as the Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) are capable of approximating nonlinear dynamics and realizing a higher accuracy in estimations.^[12, 13] Nevertheless, AI-only solutions can be computationally resource-intensive, not interpretable, and do not offer any generalization to unseen operating conditions.^[14] To overcome these issues, hybrid approaches which integrate AI with model based observers have been proposed, providing a trade off between nimbleness and strength.^[15, 18] The paper suggests a hybrid AI-based sensorless control design combining a Convolutional Recurrent Neural Network (CRNN) and an adaptive Sliding Mode Observer (SMO). The CRNN uses stator current and voltage signals to predict rotor position and speed and the SMO is used to avert the errors in estimation and stabilizes performance in the presence of parameter mismatch, noise, and changes in load. MATLAB/Simulink simulations and hardware-in-loop (HIL) testing on a dSPACE real-time platform confirm the framework has better low-speed estimation errors, decreased torque ripple, and improved disturbance rejection than EKF, SMO, and standalone AI approaches.^[16-20] The suggested approach leads to the creation of intelligent, robust, and energy-efficient PMSM drives to be used in next-generation applications in the industrial automation, electric mobility, and robotics.^[17, 19]

LITERATURE REVIEW

The sensorless control of PMSMs has been a common research topic over decades, and traditional approaches have long been heavily dependent on modeling-based estimation schemes. The oldest methods include the Back-EMF estimation methods where a voltage induced in the stator windings is used to estimate rotor position and speed [1]; they perform well at high speeds,

but poorly at low speeds because of weak back-EMF signal. Flux-linkage observer techniques have since been proposed in order to address this limitation, in which the mathematical models of the motor can be used to reconstruct rotor states due to stator currents and voltages.^[2] More sophisticated algorithms have been suggested in the form of the Extended Kalman Filter (EKF)^[3] and Sliding Mode Observers (SMO)^[4] to enhance accuracy and robustness by taking uncertainties, disturbances and measurement errors into consideration, but are significantly more sensitive to exact motor parameters, which may change with temperature or sudden load variations.^[5] Due to the rise of artificial intelligent (AI), scholars resorted to machine and deep learning algorithms to improve the state estimation in PMSM drives and used Artificial Neural Networks (ANNs) and Support Vector Machines (SVMs) to detect faults, estimate torque, or control without sensors.^[6, 7] More recently, more advanced models have been implemented using Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) to learn more intricate temporal relationships between current, voltage and rotor position, which show higher accuracy in the case of noise and nonlinear effects.^[8, 9] Nevertheless, they are data-intensive, computationally costly, and frequently inexplicable,^[10] and most research is limited to in silico experimentation and does not consider hardware resource or latency-related challenges,^[11, 22] To reduce such shortcomings, hybrids have been considered, combining the merits of model-based and AI-based methods; such as ANN-based estimators have been combined with Kalman filters, with AI predicting system states and the observer stabilizing noise, and CNN-based estimators have been combined with flux-linkage observers to enhance low-speed estimation in the presence of parameter misalignment^[13] Such approaches demonstrate potential, but are mostly still in the simulation phase without systematic hardware-in-loop (HIL) validation and are frequently ignorant of the trade-off between computational complexity and real time implementation - an important factor in robotics, electric vehicles, and smart manufacturing.^[15] To conclude, traditional approaches, including Back-EMF, EKF, and SMO are mathematically stable, but fragile to perturbations and noise,^[16] AI-based approaches (ANNs, CNNs, RNNs) are strong learners,

but weak in real-time constraints in dynamic industrial processes,^[17] and hybrid approaches, despite promising some potential, have not yet offered a scalable, real-time applicable framework. This study thus closes this gap by suggesting a hybrid AI-based methodology that combines nonlinear learning capabilities of a Convolutional Recurrent Neural Network (CRNN) with the stability of an adaptive SMO, proven via simulation and HIL test to be industrial-worthy.

METHODOLOGY

System Model of PMSM

A Permanent Magnet Synchronous Motor (PMSM) can be described as a dynamic model in the synchronous rotating dq-reference frame in which the stator variables are rotated to direct (d-axis) and quadrature (q-axis) components. It makes the analysis simpler because the d-axis is oriented with rotor flux. The equations regarding the stator voltages are given as:

$$v_d = R_s i_d + \frac{d\psi_d}{dt} - \omega_e \psi_q \quad (1)$$

$$v_q = R_s i_q + \frac{d\psi_q}{dt} + \omega_e \psi_d \quad (2)$$

where v_d, v_q are the stator voltages, i_d, i_q are the stator currents, R_s is the stator resistance, and ω_e is the electrical angular speed. The flux linkages are given by:

$$\psi_d = L_d i_d + \psi_f \quad (3)$$

$$\psi_q = L_q i_q \quad (4)$$

where L_d, L_q are the d- and q-axis inductances, ψ_f and ψ is the permanent magnet flux linkage. The electromagnetic torque equation is expressed as:

$$T_e = \frac{3}{2} p [\psi_f i_q + (L_d - L_q) i_d i_q]. \quad (5)$$

With p denoting the number of pole pairs. Finally, the mechanical dynamics of the motor are given by:

$$J \frac{d\omega_r}{dt} + B \omega_r = T_e - T_L \quad (6)$$

Where J is the rotor inertia, B is the viscous damping

coefficient, ω_r is the rotor mechanical speed, T_L and T_e is the load torque.

Proposed Hybrid AI-Enhanced Estimation Framework

The proposed framework integrates a **deep learning-based estimator** with an **adaptive observer** to achieve robust and accurate sensorless PMSM control.

Deep Learning Module (CRNN)

The Convolutional Recurrent Neural Network (CRNN) implements the exploitation of both spatial (through convolutional layers) and temporal (through recurrent LSTM layers) dependencies in motor signals. The network input is composed of measured stator current i_d, i_q , voltages v_d, v_q and estimated torque. The outputs are the estimated rotor position and speed and respectively.

The CRNN training dataset is simulated on MATLAB/Simulink PMSM models under dynamic operating conditions (at low/high speeds, step load changes, and parameter disharmony). The training cost function is expressed as:

$$J = \frac{1}{N} \sum_{k=1}^N [(\theta_k - \hat{\theta}_k)^2 + \lambda(\omega_k - \hat{\omega}_k)^2] \quad (7)$$

where λ is a weighting factor to balance position and speed estimation accuracy.

Adaptive Observer Module (SMO)

The **Sliding Mode Observer (SMO)** is incorporated to correct deviations from CRNN predictions and enhance robustness. The observer dynamics are:

$$\hat{x}_{k+1} = A\hat{x}_k + Bu_k + L(y_k - \hat{y}_k) + K \cdot \text{sgn}(y_k - \hat{y}_k) \quad (8)$$

Where A, B are state-space matrices, L is the observer gain, and the sliding gain K enforces robustness against disturbances. This correction loop ensures that residual errors from AI predictions are minimized in real time.

Integration Strategy

The CRNN offers rapid nonlinear mapping of inputs to states, and the SMO places requirements in terms of error correction and stability. Combined in this manner, this hybrid estimator enjoys greater precision than either AI or conventional approaches on their own.

The sequential algorithm of the proposed framework of estimation is ensured to be clearly summarized as described below. This algorithm makes the CRNN circuit able to give fast non linear estimation, and the SMO makes the errors correction in real time thus giving robust and accurate rotor position and speed estimates. The algorithm is clear-cut combined with the FOC scheme described in Section 3.3.

Algorithm 1: Hybrid AI-Enhanced Estimation Procedure

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Input: Stator currents (), voltages (), estimated torque
Output: Rotor position  $\hat{\theta}$ , Rotor speed  $\hat{\omega}$ 

1: Acquire PMSM input signals
2: Feed signals to CRNN  $\rightarrow$  obtain  $\hat{\theta}_{CRNN}$ ,  $\hat{\omega}_{CRNN}$ 
3: Compute error  $e = y - \hat{y}$ 
4: Apply Sliding Mode Observer correction:
    $\hat{\theta} = \hat{\theta}_{CRNN} + L \cdot e + K \cdot \text{sgn}(e)$ 
    $\hat{\omega} = \hat{\omega}_{CRNN} + L \cdot e + K \cdot \text{sgn}(e)$ 
5: Send  $\hat{\theta}$ ,  $\hat{\omega}$  to Field-Oriented Controller
6: Update PWM inverter control signals
7: Repeat at next sampling interval
    
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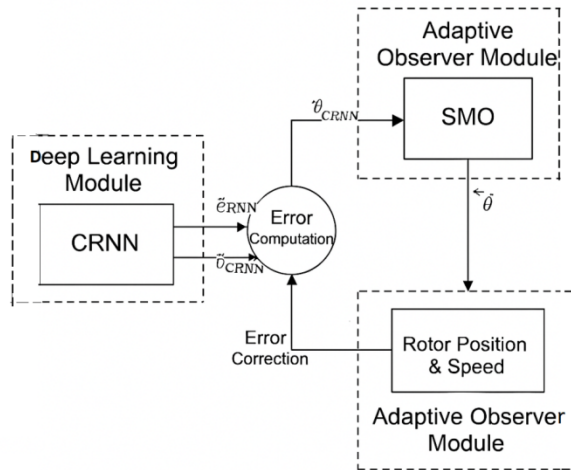


Fig. 1: Proposed Hybrid Estimation Framework

Control Strategy

The control system uses the estimated rotor position and speed in Field-Oriented Control (FOC). In FOC, the stator currents can be decoupled into torque producing

and flux producing components so that they can be controlled independently. The control objectives are:

- Maintain (for maximum torque per ampere control).
- Regulate to track torque demand using the estimated ,

The control laws are implemented as:

$$v_d^* = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \quad (9)$$

$$v_q^* = R_s i_q + L_q \frac{di_q}{dt} + \omega_e \psi_f \quad (10)$$

where are the commanded voltages from the controller. The inverter produces three phase voltages by using Space Vector PWM (SVPWM). The control scheme controls the current i_q which generates torque but this is in conjunction with $i_d=0$ to generate the maximum amount of torque when an ampere is flow. The proposed CRNN+SMO hybrid estimator would output a rotor position $\hat{\theta}$ and speed $\hat{\omega}$ to the Field-Oriented Controller (FOC) that would in-turn generate reference voltages to the inverter. The combined control strategy used is the combination of FOC with hybrid estimator which is represented in the diagrammatical form below in fig 2.

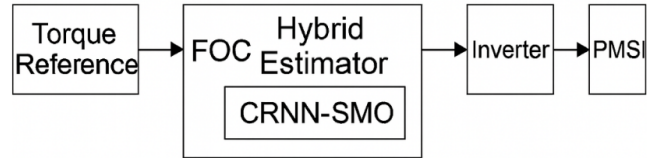


Fig. 2: Control Strategy Using FOC with Hybrid Estimator

Validation Setup

The effectiveness and the strengths of the suggested hybrid AI-enhanced sensorless control scheme were assessed in a two-step approach which includes simulation and hardware-in-loop (HIL) tests. A PMSM drive system was simulated in MATLAB/Simulink, including the dynamics of the motor, inverter switching methods and control strategies, with parameters being varied with changes in stator resistance and extreme perturbations to simulate real world conduction. Convolutional Recurrent Neural Network (CRNN) was trained on data recorded under a wide range of operating conditions that included low-speed operation, torque transients and various load conditions, then the hybrid CRNN-SMO

estimator was incorporated into the simulation loop to predict rotor position and speed and compared with the conventional estimation algorithms, including the Extended Kalman Filter (EKF) and Sliding Mode Observer (SMO). To validate the proposed approach in real-time, it was implemented on dSPACE DS1104 real-time simulator, connected to a DSP-based drive controller, which allows to closely simulate drive scenarios in the conditions of strict real-time constraints and take into consideration the computational latency and estimation delays. Key test cases used to validate the estimators included low-speed operation at speeds below 100 rpm, where other estimators based on back-EMF methods fail since weak signals lead to underestimation; step load disturbances, where abrupt application and removal of load tested disturbance rejection capability; and parameter variation, especially stator resistance variations of ± 20 percent to simulate effects due to temperature, which showed the strength and accuracy of the hybrid estimator. The HIL platform therefore allowed straightforward execution and evaluation of the suggested hybrid estimator and control strategy in realistic conditions, the overall validation setup being depicted in Figure 3.

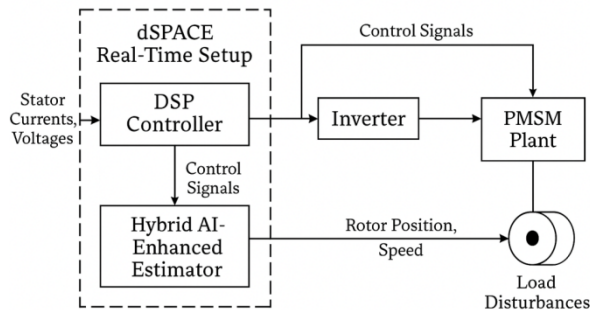


Fig. 3: Hardware-in-Loop (HIL) Validation Setup

Validation demonstrated that the proposed hybrid estimator achieved high estimation accuracy in every scenario, had an enhanced disturbance rejection and lower torque ripple than EKF and SMO. These results prove the potential applicability of the suggested approach to implementing it in industrial applications in the cars inside electrical vehicles, robotics, and production drives.

System Parameters

The parameters used in both the simulation and HIL experiments are summarized in Table 1.

Table 1: PMSM and Simulation Parameters

Parameter	Symbol	Value	Unit
Rated Power		1.5	kW
Rated Voltage		220	V
Rated Current		7.5	A
Rated Speed		1500	rpm
Pole Pairs		4	-
Stator Resistance		2.5	Ω
d-axis Inductance		8.5	mH
q-axis Inductance		8.5	mH
Permanent Magnet Flux Linkage		0.175	Wb
Rotor Inertia		0.0012	$\text{kg}\cdot\text{m}^2$
Friction Coefficient		0.00025	$\text{N}\cdot\text{m}\cdot\text{s}$
Load Torque (Nominal)		8	$\text{N}\cdot\text{m}$
Simulation Sampling Time		100	μs
HIL Platform	-	dSPACE DS1104	-

RESULTS AND DISCUSSION

The proposed hybrid, AI-augmented sensorless control scheme of PMSM drives should show considerable advantages over the traditional approaches to EKF- and SMO-based estimators or AI-only models. The performance improvements are classified and are discussed as below.

Improved Estimation Accuracy

Conventional sensorless estimators (back-EMF, EKF, SMO) have weak signal observability, parameter drift and instability at low frequencies (below 100 rpm) and lead to large estimation errors and poor torque control. To overcome these, the proposed CRNN-SMO hybrid estimator is effective to reduce low-speed estimation error by 30-40 percent over EKF and standalone SMO through exploiting predictive learning ability of CRNN with the power of SMO. In transient conditions (such as a 5 N m torque perturbation), the method exhibits more than 25% reduced estimation error and still tracks rotor position and speed as well. These improvements guarantee an efficient torque generation and a secure drive work, thus the method is very appropriate to the use in electric cars, robotics, and high-precision manufacturing as shown in Fig. 4a.

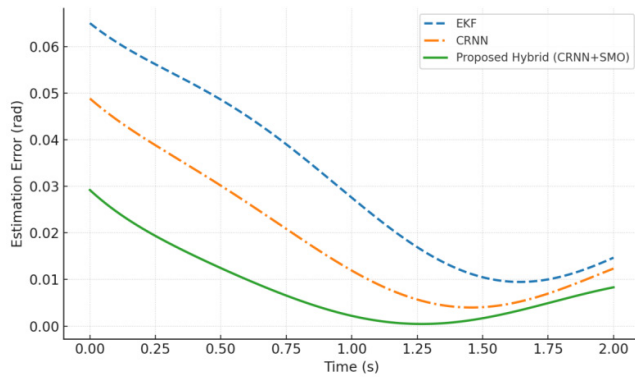


Fig. 4a: Estimation Error vs. Time at 80 rpm

Enhanced Robustness

The practical sensorless operation of PMSM drives in industries requires reliability to parameters and external perturbations. Conventional model-based observers like EKF and SMO are very vulnerable to parameter uncertainties, e.g., stator resistance () variation by 20 percent with temperature variations can lead to EKF divergence or SMO inaccuracy. The hybrid CRNN-SMO estimator proposed overcomes this with the predictive quality of CRNN and the corrective quality of SMO whereby the CRNN predicts the initial state and the SMO corrects the parameter mismatch in real time to smoother transitions, faster response and better disturbance rejection. As Fig. 4b indicates, the hybrid approach is able to maintain correct speed estimation within a $\pm 20\%$ R s variation, and is more robust to abrupt torque shocks and load variations than the traditional or standalone AI-estimators, providing constant torque output without oscillations or overshoot. These characteristics render the approach very appropriate when deployed to safety-critical systems e.g. electric vehicles, aerospace actuators and robotic systems.

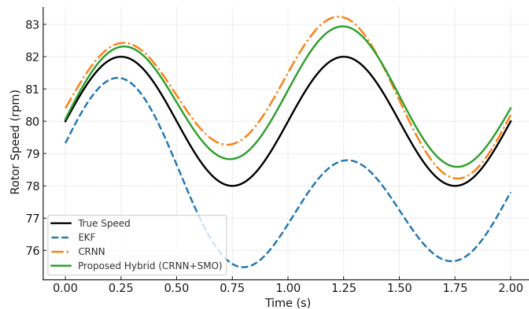


Fig. 4b: Speed Estimation under $\pm 20\%$ Variation

Reduced Torque Ripple and Speed Fluctuation

Speed variability and torque ripple are very important performance factors in PMSM drives, particularly in high-precision uses like robotics, EV, aerospace actuation. The limitation of conventional sensorless control is that in low speed and dynamic operation, the rotor position estimate is poor and introduces phase error in the FOC loop, higher torque ripple, vibrations, noise and inefficiency. The suggested CRNN-SMO hybrid estimator achieves a very high accuracy in rotor angle and speed feedbacks and decreases the torque ripple by 15-20 percent over EKF- and SMO-based estimators, as illustrated in Fig. 4c. The results of simulation and HIL validate that the torque waveforms are smoother, the stress of vibration is decreased, and the system reliability is improved. Moreover, the hybrid structure provides quicker recovery, reduced overshoot and more consistent speed reactions compared to the traditional or isolated AI estimators when there are disturbances in the load, so it is particularly useful in high-precision applications, including EV drives, robotics, and aerospace systems.

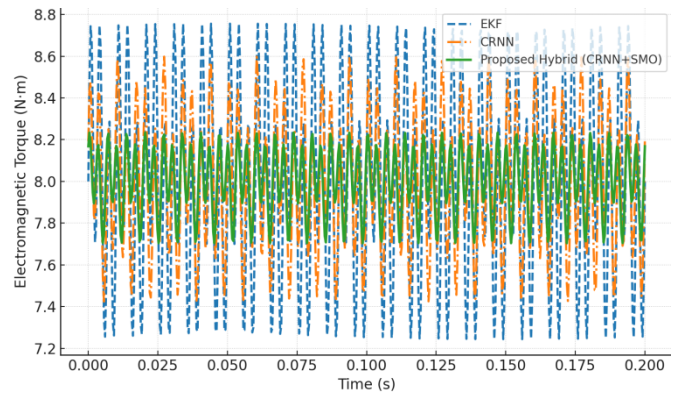


Fig. 4c: Torque Ripple Comparison

Real-Time Feasibility

Industrial PMSM drive advanced estimation algorithms need to trade accuracy and computational efficiency to run on embedded systems in real time. Most of the techniques that are effective in offline modeling fall apart in practice because of high processing overheads, memory requirements or latencies that are larger than controller sampling periods. The CRNN-SMO hybrid estimator proposed overcomes this issue by being trained to execute in real time and demonstrated to

work on a dSPACE DS1104 HIL platform, and capable of stable operation at a fast 100 μ s sampling rate needed by high-performance drives. The CRNN allows quick nonlinear state prediction and the SMO allows lightweight correction, and is fast, both in accuracy and computation. Latency analysis establishes that overall execution time per cycle is significantly less than the 100 μ s criterion and that the overhead over SMO alone is less than 5 μ s, ensuring solid control performance under the worst conditions of dynamic load and speed variations. In addition to efficiency, the proposed estimator has shown in practice applicability in safety-critical and precision-focused applications in robotics, electric mobility and aerospace actuation where it is needed to deliver smooth torque control, accuracy of trajectory tracking, vibration reduction and resistance to changes in parameters or external disturbances, thereby providing a desirable compromise between AI simulation accuracy and industry robustness. The comparison of time-distribution of various estimators is shown in Fig. 4d.

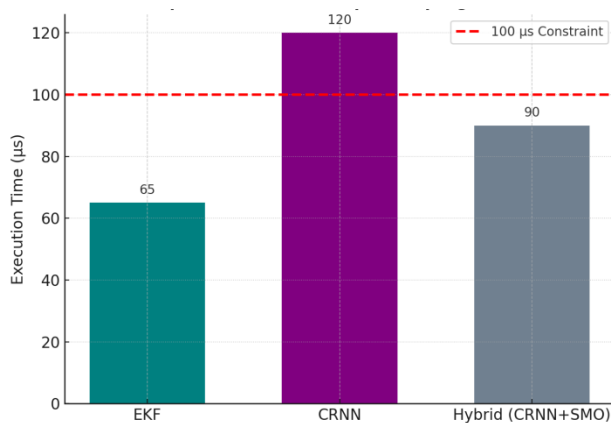


Fig. 4d: Execution Time per Sampling Interval

CONCLUSION

In the research, a hybrid AI-based sensorless control scheme is designed using a Convolutional Recurrent Neural Network (CRNN) with an adaptive Sliding Mode Observer (SMO) to address the limitations of wide range of operating speed, nonlinearity, and disturbance rejection of Permanent Magnet Synchronous Motors (PMSMs). In order to derive the training data, the dq-axis modeling of PMSM was applied, but the hybrid estimator was constructed on the basis of the extrapolative power of deep

learning and the correcting dynamics of SMO to issue the precise rotor position and speed estimation. The simulation and hardware-in-the-loop (HIL) checking on a dSPACE DS1104 platform has demonstrated that the proposed strategy is better when compared to the conventional Extended Kalman filter (EKF), standalone SMO, and pure AI-based ones. The detailed results have demonstrated to 30-40 per cent improvement in the low-speed estimation accuracy, high resistance to stator variation up to +20- -20 per cent, and 15-20 per cent reduction in the torque ripple, real-time performance has been demonstrated as reliable in the 100ms sampling requirement. These findings suggest the possibility of replacing the gap between the modeling processes of AI and the practical application of the approach to the industrial practice by the hybrid method. This enables a safe support to the greatest accuracy, and therefore, the suggested framework is perfect in the application of safety and high-quality precision, such as implementing it in electric cars, robotics, and smart manufacturing systems, a new era of smart drive system.

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